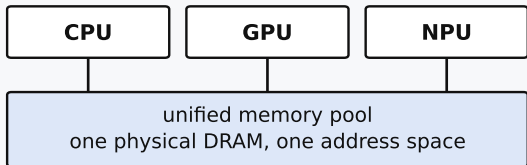


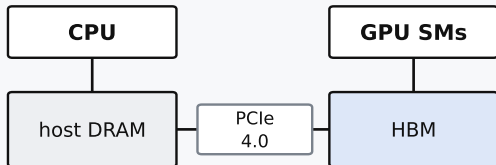
Apple silicon — unified memory



no host↔device copy — a GPU tensor is visible to NumPy

- ⇒ the randomized SVD runs in NumPy on the CPU while attention runs on the GPU — no marshalling
- ⇒ "freeing" returns memory to a shared pool, so bounding *peak* allocation is what matters

NVIDIA discrete GPU — separate HBM pool



explicit H2D / D2H — a block restore is ≈ 137 KB, ≈ 15 μ s

- ⇒ a step-ahead prefetch on a dedicated cudaStream hides the restore behind the previous decode step
- ⇒ the GPU-native Gram-Eigh SVD keeps compression on device, avoiding the D2H copy of the *K/V* block